A Novel TSV Model With Fault Characterization for High-Frequency Transmission in 3D ICs

Prosen Kirtonia, *Member, IEEE*, Shelby Williams, *Member, IEEE*, Sonia Akter, *Member, IEEE*, Magdy Bayoumi, *Life Fellow, IEEE*, and Kasem Khalil, *Senior Member, IEEE*

***Abstract*—Through-silicon vias (TSVs) are essential for 3D integrated circuits (ICs) and advanced chiplet packaging. The semiconductor industry is transitioning toward 3D ICs, chiplets, and system-in-package (SiP) solutions due to the slowdown of Moore's Law and limitations in conventional silicon scaling. In this paper, we propose an optimized TSV architecture for high-frequency transmission to enhance its suitability for 6G communication chips, and develop a comprehensive equivalent circuit model for fault-free and faulty TSVs. This model accounts for open-circuit and short-circuit fault conditions while considering the effects of higher frequencies, substrate type, doping concentration, and adjacent layers. At the physical level, the TSVs are simulated using the Ansys High-Frequency Structure Simulator (HFSS), and the equivalent circuits are designed using the Cadence Virtuoso tool. An experimental evaluation is also conducted to validate the physical design. We position the TSVs in a pattern of ground-signal-ground (G-S-G) to reduce the effective inductance of the signal TSV, thereby minimizing inductive reactance at ultra-high frequencies. Consequently, the reflection coefficient remains below -10 dB across the frequency range of 0.1 to 146.3 GHz. Furthermore, we compare simulation outcomes from HFSS and Cadence for both fault-free and faulty TSVs under varying operating conditions. Additionally, the parasitic circuit components are characterized through extensive theoretical derivations for in-depth circuit verification. Collectively, the rigorous analysis, experimental validation, and thorough investigation of the proposed design and its equivalent circuit demonstrate their potential for use in creating datasets for a fault prediction machine learning model.**

***Index Terms*—Through-silicon vias, TSV, chiplet, 3D ICs, open circuit, short circuit, HFSS, Moore's Law, ECC.**

I. Introduction

THE rapid progress in Artificial Intelligence (AI) and the Internet of Things (IoT) has led to exponential growth in annual data generation. To handle this ever-increasing volume of data, the demand for computationally powerful systems with efficient performance and compact designs has surged. Simultaneously, the diminishing returns of Moore's law and the increasing cost of advancing to smaller process nodes have imposed immense challenges on the semiconductor industry [1], [2], [3], [4]. Consequently, chiplets, SiPs, and 3D ICs have drawn significant research attention as alternatives to conventional system-on-chip (SoC) to meet the growing demand [5].

In chiplets and SiPs, advanced packaging plays a vital role in determining performance, delay, security, and compactness. The advanced packaging and 3D IC integration rely heavily on chip interconnects for enhanced and secure data transmission. Among the various interconnect technologies, TSV is one of the most prominent choices, as it enables the vertical stacking of chips while offering optimal performance [6], [7], [8]. In addition, packaging and fabrication techniques such as local backside etching, flip-chip with wafer-level Cu-pillar bumps, and backend-of-line integrated on-chip signal transition can significantly enhance the system performance. These approaches highlight the diverse innovation landscape in advanced IC packaging [9], [10], [11].

Today's AI accelerators, whether graphics processing units (GPUs), tensor processing units (TPUs), or custom application-specific integrated circuits (ASICs), commonly integrate high bandwidth memory (HBM) and SoC components via an interposer using TSVs attached to the substrate. This enables modern high-performance computing (HPC) systems [12], [13], [14]. However, this setup alone cannot meet future HPC demands, which require higher-density, low-energy computing, and increased bandwidth [15], [16]. Achieving this will involve vertical stacking of multiple silicon components and the integration of more HBM, requiring a larger interposer and extensive TSV usage [12]. Apart from this, 3D ICs for 6G communications must support operation at frequencies beyond 100 GHz. Therefore, the high-frequency signal transmission in TSVs is essential for enhancing speed, bandwidth, and power efficiency [17], [18]. In addition, research is ongoing to deploy wireless interconnects for 3D ICs to transmit high-frequency signals, which would significantly increase complexity [19]. Thus, conventional TSVs with high-frequency data transmission capability could meet the requirements while maintaining simplicity and traditional design.

Advancing TSV technology to a cutting-edge stage requires extensive research and development efforts. To accomplish

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Prosen Kirtonia and Magdy Bayoumi are with the Department of Electrical and Computer Engineering, University of Louisiana at Lafayette, Lafayette, LA 70504 USA (e-mail: Prosen.kirtonia1@louisiana.edu; magdy.bayoumi@louisiana.edu).

Shelby Williams and Sonia Akter are with the Center for Advanced Computer Studies, University of Louisiana at Lafayette, Lafayette, LA 70504 USA (e-mail: shelby.williams@louisiana.edu; sonia.akter1@louisiana.edu).

Kasem Khalil is with the Electrical and Computer Engineering Department, University of Mississippi, Oxford, MS 38677 USA, and also with the Electrical Engineering Department, Assiut University, Asyut 71515, Egypt (e-mail: kmkhalil@olemiss.edu).

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this, a highly accurate R-L-C equivalent circuit model must be developed for TSVs under various conditions, including fault-free and faulty scenarios, while considering all parasitic components, effects, and characteristics. These parasitic components of the TSV, pads, bumps, interposer, and other layers dictate the system performance [20]. Thus, well-defined modeling will enable early estimation of system performance before fabrication, augmenting flexibility in research and development of fault-tolerant systems [21], [22].

The reliability and robustness of TSVs are critical prerequisites for successful fabrication [23]. However, faults in TSVs, such as open- and short-circuit faults, are not uncommon and pose significant challenges to ensuring the overall reliability of the system. These faults may arise from manufacturing defects, which can be detected before device integration, or from aging and stresses during operation. In particular, voids or insulator breakdowns caused by thermal or mechanical stress, electromigration, and aging can lead to undetected faults in TSVs, adversely affecting chip performance and lifespan of the device [24], [25]. Therefore, detecting or predicting these faults at an early stage and in real-time can substantially enhance system reliability and performance. Additionally, implementing a self-healing technique with redundant TSVs could further improve system resilience and avoid disruptions in the operation. However, an accurate and detailed equivalent circuit model of physical TSVs plays a pivotal role in enabling this advancement [26]. The contributions of this paper are as follows:

- Fault-free TSVs in a G-S-G structure are designed using HFSS, with an optimized height-to-pitch ratio to reduce the impact of inductive reactance at higher frequencies. As a result, a simulated reflection coefficient of less than -8 dB is estimated up to 200 GHz. Additionally, an experimental analysis of the proposed architecture is carried out.
- A holistic equivalent circuit is developed for the designed TSVs, considering the impacts of the skin effect, depletion layer, temperature, pad, insulation layer, inter-metal dielectric (IMD) layer, passivation layer, and other factors.
- Short-circuit and open-circuit faults are introduced into the physical-level design for various crack lengths and locations, and their effects are analyzed.
- The equivalent circuit for faulty TSVs is also modeled extensively using Cadence Virtuoso and compared with the HFSS results.
- The equivalent circuit is validated by conducting an in-depth theoretical analysis and matching simulated results at various levels.

The remainder of the paper is organized as follows: **Section II** reviews the recent state-of-the-art literature to highlight research gaps and our key contributions. **Section III** describes the physical design of proposed TSVs in HFSS, while **Section IV** presents the equivalent circuit design along with the parametric and mathematical analysis. **Section V** discusses the faults and their equivalent circuits with detailed assessment, followed by **Section VI**, which presents the results and compares physical and circuit-level outcomes for validation. Finally, the paper is concluded in **Section VII** with a discussion and directions for future work.

II. Literature Review

Over the past decade, research on TSVs has received particular attention, especially in enhancing their reliability and improving performance in 3D ICs. To carry this out, it is important to understand, and thus vital to investigate, the electrical properties of TSVs. To that end, this literature review systematically outlines major advancements from 2013 to 2025 and identifies critical gaps in the existing TSV research field.

Rahman et al. [27] in 2013 concentrate on modeling and optimizing TSV geometries for 3D integration, utilizing 3D full-wave simulation (HFSS) and circuit simulation (ADS). Their findings reveal that cylindrical TSVs achieve minimal insertion loss and deliver superior performance compared to other structures under the same conditions. That same year, Pang et al. [28] evaluate the electrical performance of a 3×3 TSV array with diverse signal and ground patterns, proposing an optimized TSV array design and practical insights for 3D packaging. Despite these contributions, both studies are primarily limited to analyzing fundamental transmission characteristics. Later, in 2015, Zhen et al. [29] investigate the transmission properties of differential TSV structures in 3D packaging. Their findings underscore the significant impact of structural parameters on forward transmission gain, including TSV diameter, insulator thickness, and differential TSV pitch. However, their study does not incorporate fault scenarios into the design. In a separate study, Georgoulopoulos and Hatzopoulos [30] introduce a fault detection technique utilizing ring oscillators (RO) for pre-bond TSV testing. Their approach relies on identifying frequency differences between faulty and fault-free TSV circuits through an RO-based detection mechanism. This method demonstrates effectiveness in detecting faults due to manufacturing defects.

Park et al. [31] in 2017 propose the R2-TSV structure, a repairable and reliable TSV set that reutilizes redundancies to improve reliability. This approach reduces inefficiencies associated with TSV redundancies, yet it mainly focuses on the redundancy and repair aspects, leaving out electrical performance analysis and circuit modeling. In 2019, Yu et al. [32] present an improved RO-based (IRO) post-bond TSV test method specifically targeting leakage faults. Their approach also uses an equivalent circuit to identify weak leakage faults. One year later, Bae and Park [33] propose an efficient TSV fault detection scheme for high-bandwidth memory using error pattern analysis. They use an erasure-based Error Correction Code (ECC) mechanism. Their method demonstrates a high detection success rate and low overhead. However, the main goal of these works was to design a pre-bond/post-bond fault detection method using RO/pattern analysis. Wang et al. [34] also contribute by investigating the effect of thermal stress on the electrical properties of TSV inductors, yet their work focused solely on thermal stress, omitting the combined effects of other factors that TSVs might encounter in real-world applications. Entering 2021, Xu et al. [35] propose a novel pre-bond TSV test solution, incorporating an improved RO with a voltage-divider structure. Even though their work demonstrates

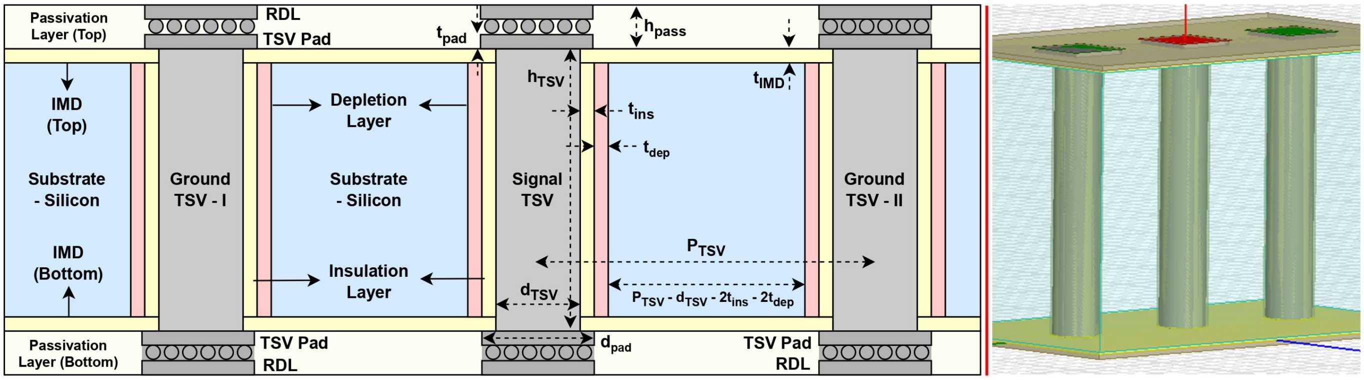


Fig. 1. Schematic of fault free TSVs in G-S-G architecture with HFSS arrangement.

greater sensitivity to TSV faults compared to traditional test methods, they do not focus on TSV characterization, which could be utilized for post-bond fault detection.

Furthermore, Wang et al. [36] investigate the transmission properties of G-S-G TSV structures at high frequencies of 60-100 GHz. They thoroughly examine how material selection and physical structure affect the electrical performance of TSVs. However, the targeted frequencies for the latest 6G technologies are more than 100 GHz. Similarly, in 2021, Yuejuan et al. [37] examine the impact of geometric dimension and insulation layers on the RF performance of truncated cone-shaped TSVs. It highlights the critical role of sidewall angle and insulation layer optimization in enhancing electrical performance. Moreover, Zeng and Zhu [38] in 2023 present an analytical method to calculate the resistance of the TSVs. Their method shows an error margin of less than 7% compared to the HFSS simulations. Its focus on air-core inductors limits its applicability when materials or structures vary. In addition, two recent works by Guan et al. [39] and Kim et al. [40] focus on optimizing TSV architecture and performance. Guan et al. present an equivalent circuit model for elliptical TSVs, showing better crosstalk optimization and transmission characteristics than traditional cylindrical TSVs. In 2025, Huang et al. [41] present equivalent circuit models and HFSS simulations for TSVs using micro-bump and hybrid bonding techniques. Their analysis demonstrates an 8% eye-opening improvement with hybrid bonding but lacks fault characterization. Wang et al. [42] introduce a pre-bond TSV detection method to identify open and leakage faults by modeling their mathematical relationship with (dis)charge time. Similarly, Dou et al. [43] propose a pulse-based pre-bond TSV testing method that employs a pMOS driver and pulse shrinking techniques to enhance fault detection resolution. Although effective, the contribution focuses mainly on designing the manufacturing fault detection technique, rather than advancing the TSV for higher frequencies or improving circuit accuracy by modeling additional factors.

Despite significant advancements and ongoing research, there remains ample scope for improvement and the introduction of novel features to enhance system performance. The studies we reviewed focus on either evaluating the electrical performance of TSVs or detecting manufacturing defects before system integration. Moreover, the existing equivalent circuit models presented in the literature are often simplified, omitting key influencing factors. To the best of our knowledge, none of these works have explored techniques aimed at upgrading TSV performance at frequencies exceeding 100 GHz for future 6G applications, while concurrently developing a fully inclusive equivalent circuit and characterizing faults that account for in-depth parameters. In contrast, we address these gaps by proposing an optimized G-S-G TSV model capable of operating up to 146.3 GHz, along with a comprehensive equivalent circuit and fault analysis. A detailed comparison table provided later in the discussion section, which summarizes the key considerations of each compared paper, further broadens the scope of the literature survey and underscores the contributions of this work.

TABLE I
TSV DESIGN SPECIFICATIONS

Parameter	Value (μm)
TSV Height, h_{tsv}	60
TSV Diameter, d_{tsv}	10
Insulator Thickness, t_{ins}	0.5
IMD Layer Thickness, t_{imd}	0.5
Passivation Layer Thickness, h_{pass}	1
Pitch	30
Pad Thickness, t_{pad}	1
Pad Diameter, d_{pad}	13

III. TSV DESIGN IN G-S-G STRUCTURE

TSVs serve as the backbone of 3D ICs for HBM, advanced packaging, and AI accelerators. In this work, we design TSVs using HFSS to study their characteristics and behavior for optimization. We chose G-S-G arrangement to reduce the impact of inductance, enabling high-frequency signal transmission. At higher frequencies, inductance becomes increasingly dominant, degrading signal quality and increasing reflections. The schematic of the proposed design is delineated in Fig. 1.

We utilize a cylindrical conductor of height h_{tsv} and diameter d_{tsv} as the core of the TSV. To isolate the conductor from

TABLE II
MATERIAL PROPERTIES

Parameter	Value
Temperature, T	298 K
Conductivity (Cu), σ_{cu}	5.8×10^7 S/m
Conductivity (Si), σ_{si}	7.68 S/m
Resistivity (Cu) @ 283K, ρ_0	1.68×10^{-8} Ω·m
Resistance Thermal Coefficient, α	0.039 /°C
Permittivity (Si), ϵ_{sub}	11.9
Permittivity (Polyimide), ϵ_{pass}	3.5
Permittivity (SiO_2), ϵ_{ins}	4
Intrinsic Carrier Concentration, n_i	1.45×10^{16} m^{-3}
Doping Concentration, N_a	10^{21} m^{-3}
Hole Mobility, μ_p	0.048 m^2/V·s

the silicon substrate, an insulating layer of thickness t_{ins} is employed around the core. In addition, a depletion layer of thickness t_{dep} is considered in the design to account for the impact of the metal-semiconductor interface. Moreover, pads with a diameter d_{pad} are added at both ends of the TSV, and the specifications of the design parameters are outlined in Table I.

In the HFSS design, we employ copper as the core conducting material, while silicon dioxide is used as both the insulating and IMD layer. P-type lightly doped silicon is chosen as the substrate layer. Additionally, polyimide is used as the passivation layer for its excellent stress buffering capability, low moisture absorption, and low dielectric constant. Consequently, the passivation layer, positioned at both ends near the chips, prevents corrosion and leakage between the chips, while also mitigating cracking and stress-related failures, and enhancing reliability. The material properties of the proposed design are summarized in Table II.

One of the key objectives of the proposed work is to introduce ultra-high-frequency signal transmission in TSVs. In general, impedance matching is crucial for efficient signal transmission. The impedance of a TSV depends on its resistance, inductance, capacitance, and operating frequency, as outlined in (1). As frequency increases, inductive reactance increases, while capacitive reactance decreases. Hence, TSVs behave more inductively at higher frequencies. An increase in frequency leads to impedance mismatch and degrades transmission quality. Therefore, the major challenge in realizing this objective is mitigating the impact of higher-frequency components on the parasitic elements of TSV. Most importantly, reducing the effective inductance of TSVs can significantly alleviate the impact of frequency on impedance. Moreover, the skin effect negatively affects transmission efficiency at higher frequencies by increasing AC resistance. Consequently, impedance and reflection increase, distorting the signal at the output end. However, the transmission quality can be improved by reducing these effects at higher frequencies.

$$
\begin{aligned}
&Z = R + jX; \quad X = X_L - X_C\\
&\text{Inductive Reactance: } X_L = 2\pi f L_{\text{eff}}\\
&\text{Capacitive Reactance: } X_C = \frac{1}{2\pi f C}\\
&\lim_{f\to\infty} X_C \approx 0 \text{ and } \lim_{f\to\infty} X_L = 2\pi f L_{\text{eff}}\\
&\text{If, } L_{\text{eff}} \approx 0, \quad \lim_{f\to\infty} X_L \approx 0 \text{ and } \lim_{f\to\infty} Z \approx R
\end{aligned} \tag{1}
$$

The G-S-G structure of TSVs in the proposed design reduces the effective inductance and facilitates the impedance to be less sensitive to inductive reactance and frequency. Inductance generates a circular magnetic field around the conductor, as Ampere's Circuital Law [44]. The direction of the magnetic field follows the right-hand rule, following Maxwell's equation and fundamental electromagnetic (EM) wave principles [45]. A conductor is impacted by two types of inductance: self-inductance and mutual inductance. Self-inductance is the property of a single conductor or coil to oppose changes in its current by generating a back electromotive force as per Lenz's law [46]. Whereas, mutual inductance occurs when a changing current in one conductor induces a voltage in another nearby conductor due to their magnetic field interaction. The direction of mutual inductance can be the opposite of self-inductance, reducing the effective inductance based on the direction of current flow and magnetic field [45].

In the proposed design, the center TSV is for signaling, and the adjacent two are referenced as ground TSVs. Thus, the direction of the current flow in the signal and ground TSVs is opposite to each other. We can consider a TSV as a straight cylindrical wire, and the entire architecture is equivalent to parallel wires. Therefore, the self and mutual inductance of TSVs can be expressed by the classical equations of (2), (3), and (4) [47]. Mutual inductance varies with the distance between TSVs, so M_1 and M_2 imply mutual inductance for near and far TSVs, respectively.

$$
\begin{aligned}
L_{\text{self}} = \frac{\mu \cdot \mu_{\text{tsv}} \cdot h_{\text{tsv}}}{2\pi} \cdot \Bigg(\ln\left(\frac{h_{\text{tsv}} + \sqrt{h_{\text{tsv}}^2 + r_{\text{tsv}}^2}}{r_{\text{tsv}}}\right) \\
- \sqrt{h_{\text{tsv}}^2 + r_{\text{tsv}}^2} + \frac{h_{\text{tsv}}}{4} + r_{\text{tsv}} \Bigg)
\end{aligned} \tag{2}
$$

$$
\begin{aligned}
M_1 = \frac{\mu \cdot \mu_{\text{tsv}} \cdot h_{\text{tsv}}}{2\pi} \cdot \Bigg(\ln\left(\frac{h_{\text{tsv}} + \sqrt{h_{\text{tsv}}^2 + \text{pitch}^2}}{\text{pitch}}\right) \\
- \sqrt{h_{\text{tsv}}^2 + \text{pitch}^2} + \frac{\text{pitch}}{h_{\text{tsv}}} \Bigg)
\end{aligned} \tag{3}
$$

$$
\begin{aligned}
M_2 = \frac{\mu \cdot \mu_{\text{tsv}} \cdot h_{\text{tsv}}}{2\pi} \cdot \Bigg(\ln\left(\frac{h_{\text{tsv}} + \sqrt{h_{\text{tsv}}^2 + (2 \cdot \text{pitch})^2}}{2 \cdot \text{pitch}}\right) \\
- \sqrt{h_{\text{tsv}}^2 + (2 \cdot \text{pitch})^2} + \frac{2 \cdot \text{pitch}}{h_{\text{tsv}}} \Bigg)
\end{aligned} \tag{4}
$$

The direction of current flow and the magnetic field define the polarity of inductance when calculating the effective inductance. For instance, ground TSV-I has self-inductance L_{self}, and mutual inductance M_1 and M_2 from the signal and ground TSV-II, respectively. In terms of direction, L_{self} and M_2 are positive while M_1 is negative due to their opposite current flow. However, we formulate the inductance matrix in (5), incorporating all inductive components based on the designed G-S-G TSV layout to define the effective inductance of each TSV. We further derive the coupling factor matrix in (6) to illustrate how mutual inductance influences the effective inductance, as the inductive reactance at high frequencies is optimized through the use of oppositely rotating mutual inductance.

$$L_{matrix} = \begin{bmatrix} L_{\text{ground-I}} \\ L_{\text{signal}} \\ L_{\text{ground-II}} \end{bmatrix} = \begin{bmatrix} L_{\text{self}} & -M_1 & M_2 \\ -M_1 & L_{\text{self}} & -M_1 \\ M_2 & -M_1 & L_{\text{self}} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad (5)$$

$$K_L = \begin{bmatrix} K_{\text{ground-I}} \\ K_{\text{signal}} \\ K_{\text{ground-II}} \end{bmatrix} = \begin{bmatrix} 0 & -\frac{M_1}{L_{\text{self}}} & \frac{M_2}{L_{\text{self}}} \\ -\frac{M_1}{L_{\text{self}}} & 0 & -\frac{M_1}{L_{\text{self}}} \\ \frac{M_2}{L_{\text{self}}} & -\frac{M_1}{L_{\text{self}}} & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad (6)$$

The inductance of a TSV in a multi-TSV framework can be either increased or decreased, depending on the structure and current flow. In the proposed design, the self-inductance of the signal TSV is reduced by the negative mutual inductance of neighboring TSVs. This reduction helps mitigate the impact of inductive reactance on the impedance to enable high-frequency signal transmission. Accordingly, (7) defines the effective inductance of the TSVs in the proposed configuration. Overall, the effective inductance of the signal TSV is reduced, while it increases for ground TSVs.

$$L_{\text{sig,eff}} = L_{\text{self}} - 2M_1; L_{\text{ground,eff}} = L_{\text{self}} - M_1 + M_2 \quad (7)$$

For further improvement, some design parameters of the architecture can be adjusted to achieve optimal performance. As in (1), the ideal requirement is $L_{\text{eff}} \approx 0$ to reach $\lim_{f\to\infty}$. However, more ground TSVs can be incorporated into the structure if the self-inductance is high and the mutual inductance is low for a higher pitch. For instance, adding two more ground TSVs, like the G-G-S-G-G structure, introduces two more opposite rotation mutual inductance M_2 for the signal TSV into the system. Consequently, an additional component of $2M_2$ will be subtracted from L_{self}. Extending this, the effective inductance for an n-TSV architecture (Signal TSV in the center) is given by the novel expression (8). Moreover, h_{tsv}, r_{tsv}, and *pitch* can be optimized to reach the required value.

$$L_{\text{sig,eff}} = L_{\text{self}} - 2 \cdot \sum_{k=1}^{\frac{n-1}{2}} M_k \quad (8)$$

where, n is an odd integer/total number of TSVs in the architecture.

For ideal case, $L_{\text{sig,eff}} \approx 0$,

$$0 = p \cdot h_{\text{tsv}} \cdot \left(\ln\left(\frac{h_{\text{tsv}} + \sqrt{h_{\text{tsv}}^2 + r_{\text{tsv}}^2}}{r_{\text{tsv}}} \right) - \sqrt{h_{\text{tsv}}^2 + r_{\text{tsv}}^2} + \frac{h_{\text{tsv}}}{4} \right.$$
$$\left. + r_{\text{tsv}} \right) - 2 \sum_{k=1}^{\frac{n-1}{2}} p \cdot h_{\text{tsv}} \cdot \left(\ln\left(\frac{h_{\text{tsv}} + \sqrt{h_{\text{tsv}}^2 + (k \cdot \text{pitch})^2}}{k \cdot \text{pitch}} \right) \right.$$
$$\left. - \sqrt{h_{\text{tsv}}^2 + (k \cdot \text{pitch})^2} + \frac{k \cdot \text{pitch}}{h_{\text{tsv}}} \right); \quad p = \frac{\mu \cdot \mu_{\text{tsv}}}{2\pi} \quad (9)$$

For the proposed design, $n = 3$,

$$0 = \operatorname{arccosh}(2 \cdot AR) - h_{\text{tsv}} \sqrt{1 + \frac{1}{4 \cdot AR^2}} + \frac{h_{\text{tsv}}}{4} + r_{\text{tsv}}$$
$$- 2 \operatorname{arccosh}(HPR) + 2 \cdot h_{\text{tsv}} \sqrt{1 + \frac{1}{HPR^2}} + \frac{1}{HPR};$$
$$\text{for, } AR \geq 0.5 \quad \text{and} \quad HPR \geq 1 \quad (10)$$

where, AR = Aspect Ratio, HPR = Height-to-pitch Ratio.

Practically, it is impossible to mitigate the inductive effect completely and to make the impedance insensitive to frequency, since it also affects the AC resistance. Considering the ideal scenario of zero effective inductance, we derive (9) for n-TSV as a function of tunable design parameters, including height, radius, and pitch. Furthermore, the formula is modified to (10) for the proposed architecture of 3-TSV, introducing two other critical design parameters of HPR and AR. However, height, radius, and pitch have a standard range of values for different applications and densities, and reaching $L_{\text{sig,eff}} = 0$ is somewhat unrealistic with those values. Tuning between parameters can reduce the inductive impact and increase the width of the -10 dB reflection bandwidth. In addition, balancing pad diameter and thickness also play a pivotal role in implementing ultra-high-frequency transmission. In the proposed design, the parameters of Table I are chosen to address the goal of reducing effective inductance and maintaining standard AR and HPR.

IV. The Proposed Equivalent Circuit Design

Another major contribution of this paper is the development of a comprehensive equivalent circuit of the designed TSV architecture. The proposed circuit encompasses the parasitic elements of the core conductor, insulating layer, depletion layer, IMD layers, passivation layers, substrate, and connection pads. The equivalent circuit of the proposed structure is illustrated in Fig. 2. The corresponding circuit components are color-coded using the same colors as in the schematic of Fig. 1 for improved traceability in circuit flow. Furthermore, the values of components are determined by applying the design specifications and material properties of Tables I and II in the following equations.

Physically, a TSV consists of a core conductive via for signal transmission, which is surrounded by an insulating layer and a depletion region, and extends through the silicon substrate. Here, three TSVs are arranged in parallel to form a G-S-G pattern. Since the core is conductive and behaves similarly to a transmission wire, it exhibits both resistive and inductive phenomena when current flows through it. Thus, the core corresponds to a resistor R_{TSV} and an inductor L_{self}. R_{TSV} is composed of DC resistance R_{dc} and AC resistance R_{ac}, due to the physical properties of the core and skin effect of higher frequencies, respectively. R_{dc} is insensitive to frequency and depends on the length, cross-sectional area, and resistivity of the core as outlined in (11) [48]. Moreover, resistivity fluctuates with temperature change, and the relationship is described in (12) [49]. In the proposed design, copper is the core material, and it has a resistivity of 1.68×10^{-8} Ωm at 20°C and a thermal coefficient of 0.039 /°C [50]. However, the calculation is performed at a temperature of 298 Kelvin.

$$R_{\text{dc}} = \frac{\rho_T \cdot h_{\text{tsv}}}{\pi \cdot \left(\frac{d_{\text{tsv}}}{2}\right)^2} \quad (11)$$

$$\rho_T = \rho_o \cdot (1 + \alpha \cdot (T - T_0)) \quad (12)$$

At higher frequencies, the AC resistance R_{ac} becomes dominant as the current mostly flows along the conductor's

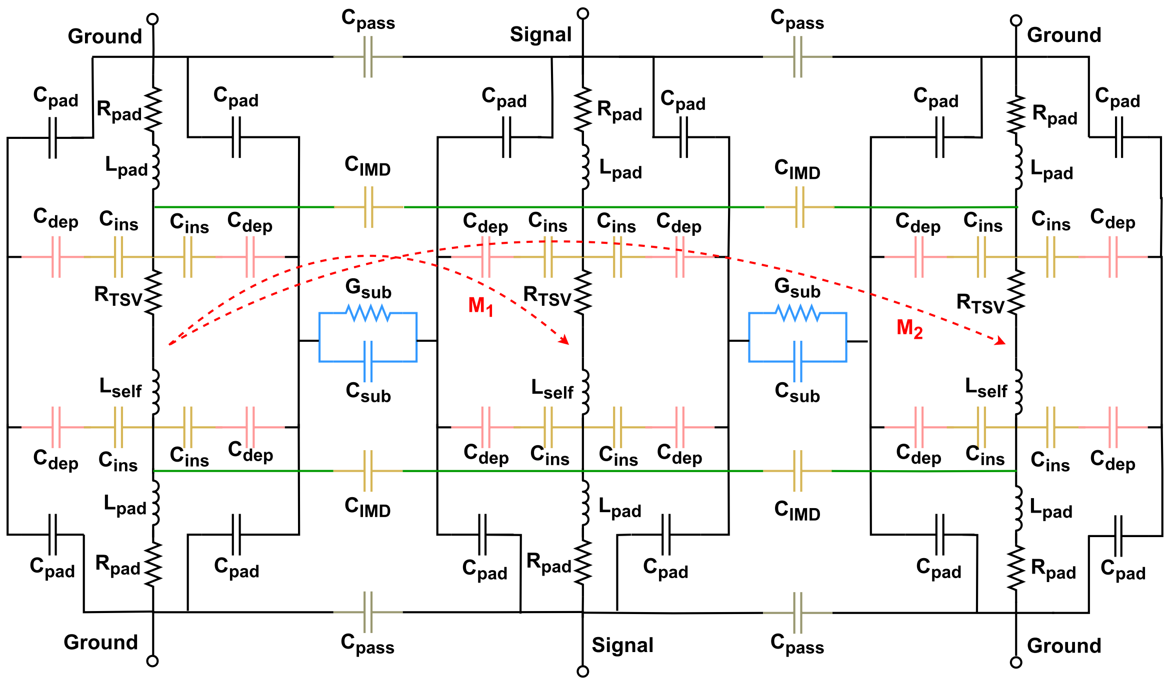


Fig. 2. Equivalent circuit of fault free TSV.

surface due to the skin effect. This effect increases AC resistance as the effective conducting area is reduced, depending on the skin depth. Skin depth can be measured using (13), which is inversely proportional to frequency [51]. As frequency increases, the alternating current concentrates near the surface of the conductor, decreasing the depth. Therefore, this reduction causes an increase in R_{ac}, according to (14) [52]. Additionally, AC resistance and skin depth are both functions of resistivity, which is sensitive to temperature. In summary, the TSV resistance defined in (15) incorporates both DC and AC components to account for the impact of higher frequencies and temperature [53]. The connecting pad resistance can be characterized and derived using the same equations as TSVs, due to their similar properties.

$$\delta = \sqrt{\frac{\rho_T}{\pi \cdot f \cdot \mu_{\text{tsv}}}} \tag{13}$$

$$R_{\text{ac}} = \frac{\text{pitch}}{d_{\text{tsv}}} \cdot \frac{\rho_T \cdot h_{\text{tsv}}}{(2\pi \cdot r_{\text{tsv}} \cdot \delta) - (\pi \cdot \delta^2)} \tag{14}$$

$$R_{\text{TSV}} = \sqrt{R_{\text{ac}}^2 + R_{\text{dc}}^2} \tag{15}$$

Inductive reactance becomes prevalent at higher frequencies, making inductors crucial components in the model. The self-inductance L_{self} due to the current flow through the conductive core and mutual inductance M_n from the adjacent TSVs play a key role in determining the performance of the model. A brief explanation of their working principle and optimization for ultra-high frequencies is given in Section III. However, their values are determined for circuit simulation utilizing (2), (3), and (4) [47].

The layer outside the core is made of silica, which serves as the insulating layer, separating the conducting core from the silicon substrate. As a result, the structure functions like two conductors separated by an insulating layer, which corresponds to a parallel plate capacitor. Particularly, it functions as a wire parallel to the wall capacitor C_{ins}, connecting the substrate and the TSV core. It can be further divided into two parallel components based on the physical architecture, and can be defined by (16) [54], [55]. Furthermore, a depletion region is formed in the substrate surrounding the insulation layer, due to the formed Schottky barrier for the metal-semiconductor interface [56]. Consequently, the depletion layer creates a similar capacitor C_{dep}, like C_{ins}, which can be determined by (17) [55]. However, C_{dep} depends on the width of the depletion layer t_{dep}, which is a function of thermal voltage, doping concentration, and intrinsic carrier concentration. Hence, we consider the lightly doped p-type silicon substrate, and the properties are showcased in Table II.

$$C_{\text{ins}} = \frac{2\pi \cdot \epsilon \cdot \epsilon_{\text{ins}} \cdot (h_{\text{tsv}} - 2 \cdot t_{\text{imd}})}{\operatorname{arccosh}\left(\frac{r_{\text{tsv}}+t_{\text{ins}}}{r_{\text{tsv}}}\right)} \tag{16}$$

$$C_{\text{dep}} = \frac{2\pi \cdot \epsilon \cdot \epsilon_{\text{sub}} \cdot (h_{\text{tsv}} - 2 \cdot t_{\text{imd}})}{\operatorname{arccosh}\left(\frac{r_{\text{tsv}}+t_{\text{ins}}+t_{\text{dep}}}{r_{\text{tsv}}+t_{\text{ins}}}\right)} \tag{17}$$

$$t_{\text{dep}} = \sqrt{\frac{4 \cdot \epsilon \cdot \epsilon_{\text{sub}} \cdot V_t \cdot \ln\left(\frac{N_a}{n_i}\right)}{e \cdot N_a}}; \text{ where, } V_t = \frac{kT}{e} \tag{18}$$

There are two additional layers in the design that correspond to capacitive properties. The IMD layers serve as insulators, and the passivation layers provide improved stress handling. The IMD separates the core of two parallel TSVs, representing the shape of two cylindrical conductors with a dielectric in between. Thus, the IMD, connecting two cores, corresponds to a parallel wire capacitor C_{imd}. C_{imd} connects two TSV lines in the circuit rather than the substrate. Furthermore, the passivation layer separates the connection pads, forming a similar capacitor, C_{pass}. In addition to other capacitors, the pad also creates a capacitor C_{pad} with the substrate, as they are also separated by dielectric material. C_{pad} connects the pad and the substrate in the model. The mathematical equations for these capacitors are outlined in (19) and (20), respectively [55].

$$C_{\text{imd}} = \frac{\pi \cdot \epsilon \cdot \epsilon_{\text{ins}} \cdot t_{\text{imd}}}{\operatorname{arccosh}\left(\frac{\text{pitch}}{d_{\text{tsv}}}\right)}; C_{\text{pass}} = \frac{\pi \cdot \epsilon \cdot \epsilon_{\text{pass}} \cdot h_{\text{pass}}}{\operatorname{arccosh}\left(\frac{\text{pitch}}{d_{\text{pad}}}\right)} \tag{19}$$

$$C_{\text{pad}} = \frac{\epsilon \cdot \epsilon_{\text{ins}} \cdot (\pi \cdot r_{\text{pad}}^2 - \pi \cdot (r_{\text{tsv}} + t_{\text{ins}})^2)}{t_{\text{imd}}} \tag{20}$$

TSV runs through the silicon substrate and bridges the stacked chips. Hence, the impact of the substrate on the TSV performance is significant. Accordingly, consideration of parasitic components of the substrate is essential for accurate circuit design. The substrate can be modeled using a conductance G_{sub} and a capacitance C_{sub}. As the substrate is lightly doped p-type silicon, the conductivity of the silicon can be determined by (21) using the acceptor concentration and the hole mobility given in Table II. Moreover, C_{sub} is a wall-to-wire parallel capacitor. However, their values are evaluated using (22) and (23) [55], [57]

$$\sigma_{si} = e \cdot N_a \cdot \mu_p \tag{21}$$

$$G_{\text{sub}} = \frac{2\pi \cdot \sigma_{si} \cdot (h_{\text{tsv}} - 2 \cdot t_{\text{imd}})}{\ln\left(\frac{\text{pitch}}{d_{\text{tsv}} + 2 \cdot t_{\text{ins}} + 2 \cdot t_{\text{dep}}}\right)} \tag{22}$$

$$C_{\text{sub}} = \frac{2\pi \cdot \epsilon \cdot \epsilon_{\text{sub}} \cdot (h_{\text{tsv}} - 2 \cdot t_{\text{imd}})}{\ln\left(\frac{\text{pitch}}{d_{\text{tsv}} + 2 \cdot t_{\text{ins}} + 2 \cdot t_{\text{dep}}}\right)} \tag{23}$$

Overall, the proposed architecture consists of multiple layers that affect performance. The equivalent components for all layers are analyzed and integrated into the proposed model. Moreover, we quantify the value of these components through detailed mathematical computations, without neglecting any parameter for simplification, to achieve maximum accuracy.

V. Fault Characterization and Circuit Modeling

Faults are common in TSVs, either due to manufacturing defects or aging. Faults caused by manufacturing defects can be detected before integration into devices [35]. However, faults due to aging are uncertain and degrade system performance, potentially leading to complete failure. Although there are many techniques to recover degraded signals from faulty TSVs at the manufacturing and testing phase, they do not apply to aging faults, as they occur during real-time chip operation in stacked configurations [58]. To address these issues, a self-healing technique with built-in redundant TSVs and a fault prediction system can be employed. However, a suitable prediction model requires an accurate and well-defined dataset to train. The motivation behind developing this detailed equivalent circuit for a wide range of frequencies, considering factors such as temperature, doping level, and hole mobility, is to characterize TSVs and their faults with higher precision to create a dataset.

A. Open-Circuit Fault

Voids or cracks in the core conductor cause open-circuit faults in TSVs. Over time, physical degradation, thermal and mechanical stress, electromigration, and other factors can break or fracture the copper core, eventually leading to a partially open fault or complete failure [59], [60]. Pre-built-in-self-tests (BIST) can detect open faults due to manufacturing defects before integrating them into final devices. However, post-BIST interrupts system performance and acts as hardware maintenance to detect ruptures caused by usage [21], [61].

Open faults can appear at the end or in between the ends of TSVs. The schematic of a TSV with a void is illustrated in Fig. 3. The length and width of the void are L_v and W_v, respectively. We introduce the void into the fault-free TSV design in HFSS for evaluating the performance deviation. In addition, faulty TSV performance is assessed for different positions (such as at the end or in between) and widths of the void, ranging from 1 to 7 μm.

We characterize the fault in the equivalent circuit and modify the values of the components affected by the fault for further analysis using the Cadence tool. Most importantly, the void increases the TSV resistance as it blocks the flow of charge. For the void in between, we divide the TSV into three parts (two parts are fault-free and one is faulty) for an accurate calculation. The lengths of the fault-free upper and lower segments are L_1 and L_2, and their cross-sectional areas are the same as in fault-free TSVs, as shown in Fig. 3. However, the cross-sectional area of the faulty zone decreases with increasing void width, raising the TSV resistance, which eventually reaches infinity for a complete fault. The new TSV resistance R_{tnb} is expressed by (24) and (26). Therefore, the higher resistance in the void causes hot spots. The void at the end impacts performance similarly, but there is one faulty zone and one fault-free zone. The resistance R_{tne} for a void at the end is outlined in (25) and (26) [51], [52], [53].

Partial open faults only affect the resistance. However, a parallel plate capacitor C_{void} is formed if the void size increases approximately to the TSV diameter. In this case, the void functions as a dielectric layer between parallel plates, where fault-free TSV zones (when the void is in between) or the pad and a fault-free TSV zone (when the void is at the end) act as parallel plates. The equivalent circuit of the faulty TSV of Fig. 3 is illustrated in Fig. 4. Moreover, C_{void} can be quantified by (27) [54].

$$R_{\text{tnb}} = \begin{cases} \infty, & \text{if } \frac{W_v}{2} \geq r_{\text{tsv}} \\ \dfrac{\rho_T \cdot L_1}{\pi r_{\text{tsv}}^2} + \dfrac{\rho_T \cdot L_v}{\pi r_{\text{tsv}}^2 - W_v^2} + \dfrac{\rho_T \cdot L_2}{\pi r_{\text{tsv}}^2}, & \text{otherwise} \end{cases} \tag{24}$$

$$R_{\text{tne}} = \begin{cases} \infty, & \text{if } \frac{W_v}{2} \geq r_{\text{tsv}} \\ \dfrac{\rho_T \cdot (h_{\text{tsv}} - L_v)}{\pi r_{\text{tsv}}^2} + \dfrac{\rho_T \cdot L_3}{\pi r_{\text{tsv}}^2 - W_v^2}, & \text{otherwise} \end{cases} \tag{25}$$

$$R_{\text{TN}} = \begin{cases} R_{\text{tne}}, & \text{if the void is at the end} \\ R_{\text{tnb}}, & \text{if the void is in between} \end{cases} \tag{26}$$

$$C_{\text{void}} = \begin{cases} \dfrac{W_v^2 \cdot \epsilon}{L_3}, & \text{if } \frac{W_v}{2} \approx r_{\text{TSV}} \\ 0, & \text{otherwise} \end{cases} \tag{27}$$

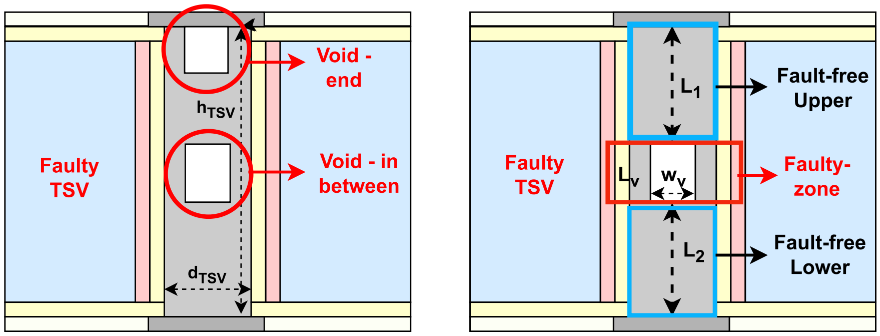


Fig. 3. Schematic of TSV with open fault.

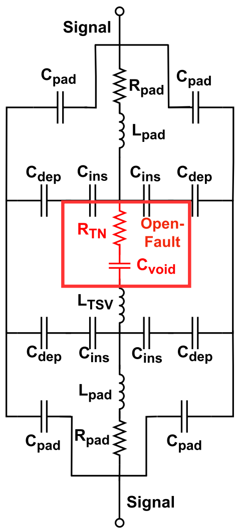


Fig. 4. Equivalent circuit of TSV with open fault.

B. Short-Circuit Fault

A fracture in the insulating layer creates a leakage path between the core and the substrate. This dielectric breakdown leads to the short circuit fault, allowing an unwanted current flow from the TSV toward the substrate. Thus, we break the insulating layer of the designed TSV in HFSS to implement the short fault and to identify the anomaly in performance [62]. The schematic of the TSV with the short fault is delineated in Fig. 5. Similar to open faults, dielectric breakdown can occur at the ends or in between. However, we evaluate the defective TSV for various locations and lengths (L_s) of the fissure.

To model the circuit of the short fault, we modify the equivalent circuit of Fig. 2 concerning the impact of the short circuit. First, it affects the capacitance of the insulating and depletion layers, C_{ins} and C_{dep}, as they break into multiple portions. Second, the short circuit will create a low-resistive path R_{sht}, connecting the core and the substrate, including an inductor L_{sht}. As shown in the schematic, the crack of length L_s in between divides the insulating and depletion layers into two parts of length L_5 and L_6. Hence, the plate length of capacitors of (16) and (17) changes, and the new values of capacitors, C_{ins1}, C_{ins2}, C_{dep1}, and C_{dep2}, can be determined by (28) and (29) [55].

$$C_{\text{ins1}} = \frac{2\pi \cdot \epsilon \cdot \epsilon_{\text{ins}} \cdot L_5}{\ln\left(\frac{r_{\text{tsv}}+t_{\text{ins}}}{r_{\text{tsv}}}\right)}; C_{\text{ins2}} = \frac{2\pi \cdot \epsilon \cdot \epsilon_{\text{ins}} \cdot L_6}{\ln\left(\frac{r_{\text{tsv}}+t_{\text{ins}}}{r_{\text{tsv}}}\right)} \tag{28}$$

$$C_{\text{dep1}} = \frac{2\pi \cdot \epsilon \cdot \epsilon_{\text{sub}} \cdot L_5}{\ln\left(\frac{r_{\text{tsv}}+t_{\text{ins}}+t_{\text{dep}}}{r_{\text{tsv}}+t_{\text{ins}}}\right)}; C_{\text{dep2}} = \frac{2\pi \cdot \epsilon \cdot \epsilon_{\text{sub}} \cdot L_6}{\ln\left(\frac{r_{\text{tsv}}+t_{\text{ins}}+t_{\text{dep}}}{r_{\text{tsv}}+t_{\text{ins}}}\right)} \tag{29}$$

Likewise, dielectric breakdown at the end has an identical impact, except that it reduces the length of the silica layer instead of splitting it into two parts. However, the changed capacitance for this updated fault position can be defined by (30) and (31) [55]. Moreover, the copper in the void of the silica layer creates an electric path to silicon. Therefore, it is equivalent to a small resistor R_{sht} and an inductor L_{sht}. R_{sht} and L_{sht} can be measured by (32) and (33) [47], [48]. However, the corresponding circuit of the TSV with short-circuit fault is depicted in Fig. 6.

$$C_{\text{insn}} = \frac{2\pi \cdot \epsilon \cdot \epsilon_{\text{ins}} \cdot (h_{\text{tsv}} - t_{\text{imd}} - L_s)}{\ln\left(\frac{r_{\text{tsv}}+t_{\text{ins}}}{r_{\text{tsv}}}\right)} \tag{30}$$

$$C_{\text{depn}} = \frac{2\pi \cdot \epsilon \cdot \epsilon_{\text{sub}} \cdot (h_{\text{tsv}} - t_{\text{imd}} - L_s)}{\ln\left(\frac{r_{\text{tsv}}+t_{\text{ins}}+t_{\text{dep}}}{r_{\text{tsv}}+t_{\text{ins}}}\right)} \tag{31}$$

$$R_{\text{sht}} = \frac{\rho_T \cdot t_{\text{ins}}}{L_s^2} \tag{32}$$

$$L_{\text{sht}} = \frac{\mu \cdot \mu_{\text{tsv}} \cdot t_{\text{ins}}}{2\pi} \cdot \left(\operatorname{arccosh}\left(\frac{t_{\text{ins}}}{\sqrt{L_s^2 + t_{\text{ins}}^2}}\right) - \sqrt{2 \cdot t_{\text{ins}}^2 + L_s^2} + \frac{t_{\text{ins}}}{4} + \sqrt{L_s^2 + t_{\text{ins}}^2}\right) \tag{33}$$

VI. Simulation and Experimental Outcomes

It is essential to validate the contribution of the proposed design and the equivalency of the designed circuits. Therefore, this section analyzes and compares the simulated and experimental findings of physical design and circuits. We evaluate reflection and transmission coefficients for different scenarios over a wide range of frequencies to identify impedance mismatch and signal integrity. Reflection coefficient (S_{11}) analysis defines the amount of signal reflected to the transmitting end due to impedance mismatch, while transmission coefficient (S_{12}) provides the proportion of signal reaching the receiving end. However, $S_{11} \leq -10$ dB and $S_{12} \geq -3$ dB are preferred for excellent transmission.

A. Fault-Free TSV

The proposed architecture of Fig. 1 is optimized to reduce the impact of inductance on the impedance to enable

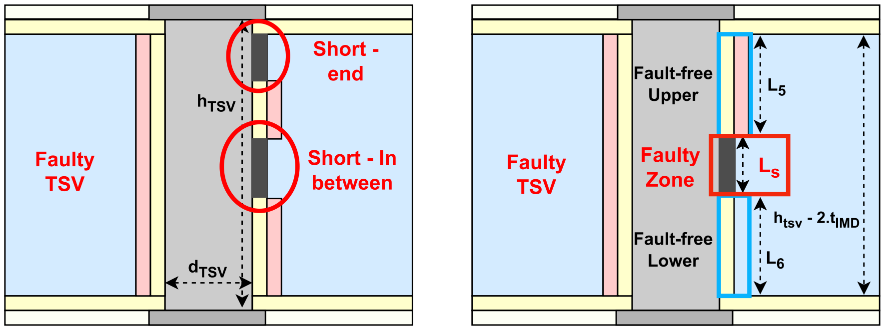


Fig. 5. Schematic of TSV with short fault.

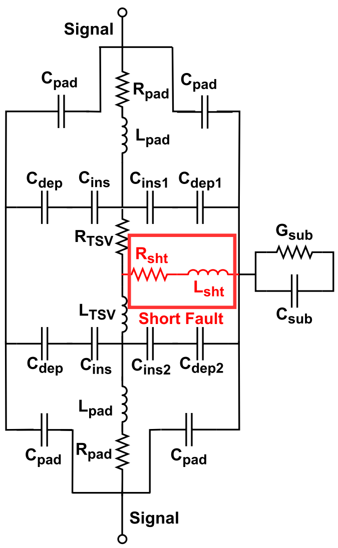


Fig. 6. Equivalent circuit of TSV with short fault.

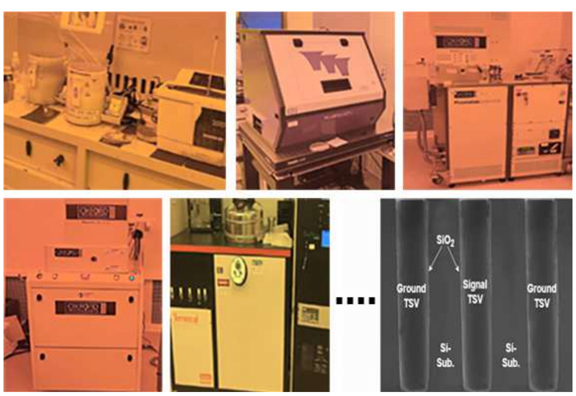


Fig. 7. Experimental setups and fabricated fault-free G-S-G TSV architecture.

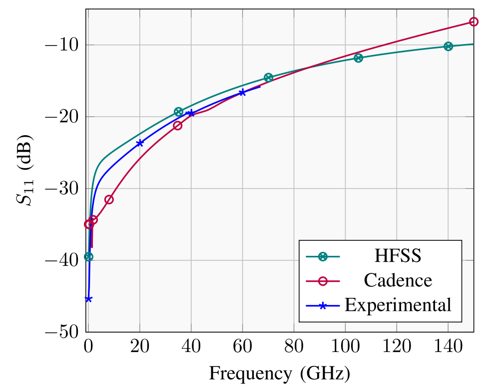


Fig. 8. Reflection coefficient comparison of physical TSV and equivalent circuit.

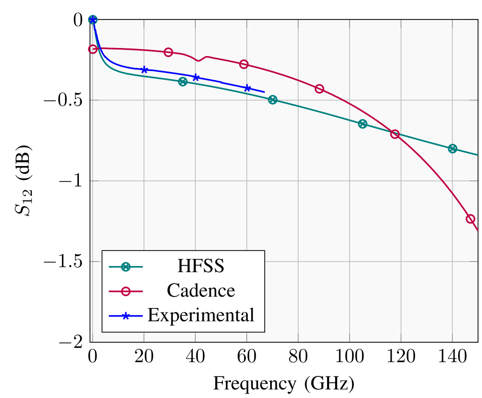
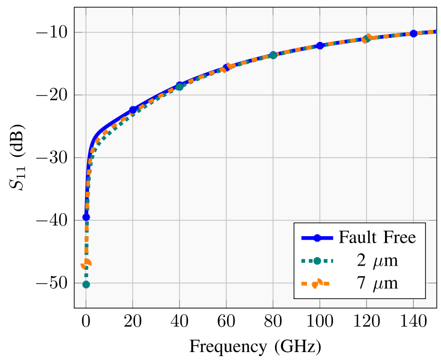


Fig. 9. Transmission coefficient comparison of physical TSV and equivalent circuit.

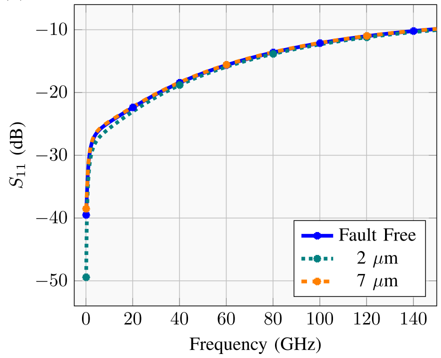
ultra-wide bandwidth and high-frequency signal transmission through the TSV. The design is simulated at the material level using HFSS, and fabricated following the setups, as visualized in Fig. 7.

The fabrication of TSVs includes a sequence of multiple precise steps. The process starts with wafer cleaning to remove residues and particles, followed by photolithography to define the TSV locations, which is illustrated in the first two images of Fig. 7. Deep Reactive Ion etching is utilized to etch the hole for TSV filling into the substrate. To ensure isolation between silicon and copper, holes are lined with silica through the Plasma-Enhanced Chemical Vapor Deposition (PECVD) process. These steps are outlined in the next three images. After deposition of the barrier and seed layer, copper is electroplated to fill the hole and create the TSVs, and excess metal is removed by polishing. The G-S-G TSV stack is depicted in the last image of Fig. 7, and the performance is measured employing a vector network analyzer (VNA) after integrating the probe pads. The simulation and experimental results are delineated in Figs. 8 and 9.

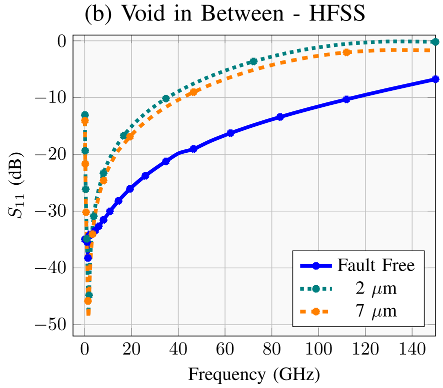
Fig. 8 illustrates the reflection coefficient, which remains less than -10 dB over the frequency range of 0.1 to 146.3 GHz in simulation. The performance metrics are evaluated at the



(a) Void of Different Sizes at the End - HFSS



(b) Void in Between - HFSS



(c) Circuit-level Output - Cadence

Fig. 10. HFSS vs. cadence simulation comparison for TSV open-circuit faults of different sizes and positions.

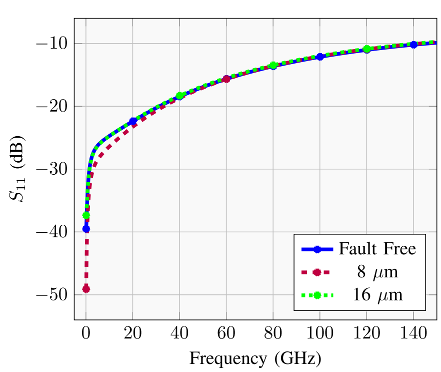


Fig. 11. Short of various lengths at end - HFSS.

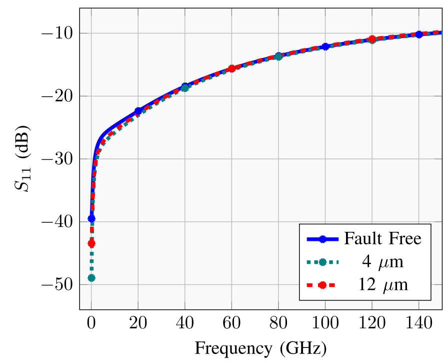


Fig. 12. Short in between - HFSS.

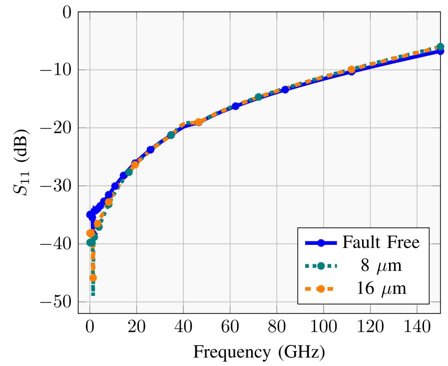


Fig. 13. Short of various lengths at end - cadence.

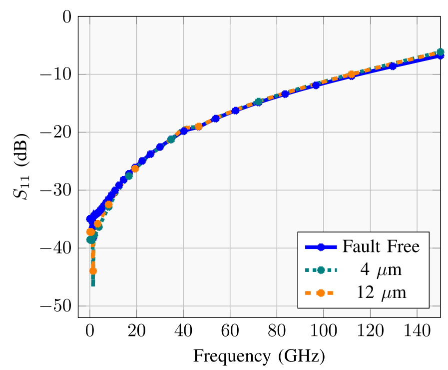


Fig. 14. Short in between - cadence.

ports of the signal TSV, which are actively excited, while the adjacent ground TSVs carry return current to emulate realistic signal propagation conditions. Thus, the current in the signal and ground TSVs flows in opposite directions, and ports 1 and 2 correspond to the input and output of the signal TSV, respectively. In the experiment, we measure S_{11} and S_{12} up to 67 GHz according to the maximum limit of the VNA. The experimental finding closely matches the simulation and is estimated to have a wider -10 dB reflection bandwidth, similar to the HFSS and Cadence results. Thus, it exhibits minimal reflection of the transmitted signal. In addition, the transmission coefficient of Fig. 9 is greater than -1 dB over the simulated frequency of 0.1 to 150 GHz, indicating minimal loss while transmitting. However, these outcomes validate the claim of ultra-high-frequency suitability of the proposed framework for 6G architectures.

The equivalent circuit designed in Fig. 2 is simulated using Cadence Virtuoso tools, and a similar parametric evaluation is performed for comparison. The reflection and transmission

TABLE III
COMPARISON OF PROPOSED WORK WITH THE STATE OF THE ART

Ref	Frequency	Fault Modeling	Eq. Circuit Modeling	Key Considerations	Structure
Guan ([39])	0.01–50 GHz	No	Yes	Elliptical TSV	S-G
Kim ([40])	0.01–20 GHz	No	Yes	IMD, underfill and pad layer; Time domain analysis; Differential signaling	G-S-S-G
Huang ([41])	1–15 GHz	No	Yes	Micro-bump and hybrid bonding; Pad, IMD and passivation layer; Time domain analysis	S-G
Zeng ([38])	70–130 GHz	No	No	Air-core TSV based inductor	–
Jung ([63])	0.01–20 GHz	Open fault, Short fault	Yes	Eye diagram analysis	G-S-G
Wang ([36])	60–100 GHz	No	No	Different TSV shapes	G-S-G
Li ([64])	1–100 GHz	No	Yes	Bumpless TSV; IMD and underfill layer	G-S
Kirtonia ([65])	0.01-30 GHz	Short Fault, Aging Fault	No	Parametric variation; Crosstalk analysis	S
Fang ([66])	0.01–5 GHz	Open fault, Short fault	No	Parametric changes with frequency	S
Hao ([60])	–	Open fault, Leakage fault	Simplistic	Fault coexistence; TSV testing	S
Niu ([67])	0.01–40 GHz	No	Yes	Capacitance characteristics; LC virtual separation	–
Liu ([22])	–	Open fault, Leakage fault	No	Fault detection through voltage and resistance	–
Wang ([21])	–	Open fault, Short fault	No	Fault detection through delay	–
This work	0.01–146.3 GHz	Open fault, Short fault	Yes	Depletion, IMD, pad, and passivation layer; Inductance optimization for ultra-high frequency; Substrate type and doping concentration; high-frequency effects	G-S-G

coefficients of the circuit are compared with the results of the experiment and the HFSS simulation in Figs. 8 and 9. The proposed circuit demonstrates a similar trend in output, with negligible differences compared to the structure-level findings of the TSV. The maximum deviation is found at 1.4 GHz between the HFSS and Cadence plots, in particular, a 0.0668% difference in power reflection. Furthermore, the reflection and transmission coefficients over the frequency range of 0.1 to 150 GHz remain below -7 dB and above -1.5 dB, respectively. The near-perfect match between the HFSS, Cadence, and measured results confirms the equivalency of the proposed circuit, including its high-frequency transmission capability.

B. TSV With Open-Circuit Fault

The void is made in the TSV to introduce the open-circuit faults, as shown in Fig. 3. We simulate the faulty TSV for varying locations, such as at the end and in between, and for the void lengths of 1 to 7 μm. However, the findings of the HFSS simulation for the void length of 2 and 7 μm with varied positions are depicted in Figs. 10a and 10b, compared to the result of fault-free TSV.

Partial open faults exhibit similar performance regardless of their position and length. S_{11} decreases at lower frequencies, indicating a lower reflection, but the loss increases due to localized heating and an increase in resistance. At higher frequencies, the skin effect dominates in the conductor. As described in (13), skin depth decreases with increasing frequency. Therefore, the most current flows near the surface rather than through the core, which reduces the influence of core-related parasitic components compared to lower frequencies. Consequently, it is more effective to test for partial open faults at lower frequencies, preferably in the range of 0.01 to 40 GHz.

The circuit of the TSV with an open fault, displayed in Fig. 4, is simulated to identify the performance deviation due to the fault and validate its correspondence to physical design. The void size changes the resistance consistently, whether it is at the end or in the middle, as the cumulative length of faulty and non-faulty zones remains the same. Consequently, the simulated output for changing void locations remains the same. We calculate the increase in resistance for the void size of 1 to 7 μm and evaluate the circuit. Fig. 10c illustrates the reflection coefficient of the circuit for void lengths of 2 and 7 μm. However, it shows a response comparable to the physical simulation. At lower frequencies, the reflection decreases, but increases at high frequencies. Together, graphs of HFSS and Cadence demonstrate matched trends, confirming the circuit's ability to represent the open-circuit fault of physical TSV.

C. TSV With Short-Circuit Fault

The schematic of the short-fault TSV is illustrated in Fig. 6. The design is simulated similarly in HFSS for two different locations and various lengths, ranging from 4 to 20 μm of insulator breakdown. The outcomes are analogous to the open-circuit fault. Reflection to the generating end is reduced, but the fault produces a low-resistive path to the substrate, causing unwanted current flow toward the silicon rather than the receiving end. Therefore, power absorption increases, which converts to heat dissipation and thermal stress. However, Figs. 11 and 12 present the computational outcomes of the physical design.

We then simulate the designed circuit with a short fault. Following the equations in Section V-B, the values of the circuit components are measured for various lengths and locations and simulated accordingly. The outcomes for different positions and lengths of faults at the circuit level are depicted in Figs. 13 and 14. The circuit also yields a comparable response to the material-level output, precisely representing short faults of TSV.

VII. Discussion and Conclusion

The novelty of this paper includes the increased capability of high-frequency signal handling of the TSV for 6G Hardware, minimally affecting the impedance by optimizing the inductance domination. In addition, the paper presents the comprehensive and precise equivalent circuits for fault-free and faulty TSVs, including the impact of higher frequencies such as skin effect, well-specified substrate, semiconductor-metal interface, and temperature. However, the results and characteristics of the proposed work are compared with recent similar works in Table III, demonstrating the advancements and the broader scope of our work. The key motivation for this work is to document a large dataset from the circuit- and physical-level TSV design in various scenarios and faults, which can be utilized further in fault prediction systems.

TSVs are substantial for 3D ICs, which are going mainstream with HBM, AI accelerators, and HPC/cloud servers, far exceeding reticle limits and extending beyond 2.5D integration. Hence, the chip's performance is highly dependent on TSVs. Manufacturing defects in TSVs can often be detected before they are incorporated into the device. However, post-integration fault detection and correction remain a significant challenge, as no existing technique can identify and resolve such issues without incurring system downtime, to the best of our knowledge. Thus, a prediction model with a self-healing system by adding optimized redundant TSVs has the potential to address the aforementioned challenge and maintain efficiency longer than before.

The near-identical outcomes of HFSS, experiment, and Cadence at the material and circuit level demonstrate the higher accuracy of the circuit in presenting the TSV and its faults. Moreover, the designs are reconfigurable as all the possible variables and their standard values are considered in the calculation instead of neglecting them to simplify the computation. Consequently, architecture and circuits can easily be modified for distinct materials and environments with different properties. However, the circuit model is less sensitive to capturing tiny anomalies in the physical model at ultra-high frequencies, as the skin effect dominates, and a couple of additional factors need to be incorporated into the design. Moreover, thermal considerations in the current model are limited to some circuit components and are not as robust as the stress reduction approach implemented at the physical level using a polymer passivation layer. Collectively, the optimized TSVS design in G-S-G layout, detailed circuit design, and fault characterization validate its competence for the creation of a dataset. In future work, thermal and stress effects on the TSV model will be explored, including variations in the TSV shapes, before starting data collection for the fault prediction model.

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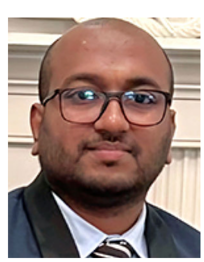


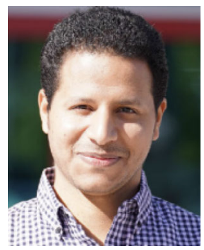
Prosen Kirtonia (Member, IEEE) received the B.Sc. degree in electronics and telecommunication engineering from Rajshahi University of Engineering and Technology, Bangladesh, in 2018, and the M.Sc. degree in electrical engineering from the University of Louisiana at Lafayette, USA, in 2025. He is currently pursuing the Ph.D. degree in system engineering with a concentration in electrical engineering with the University of Louisiana at Lafayette. His research interests include VLSI, microelectronics, 3D-ICs, chiplets, and TSVs.

Shelby Williams (Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering and computer science from the University of Louisiana at Lafayette, Lafayette, LA, USA, in 1996 and 2020, respectively, and the Ph.D. degree in computer engineering from the University of Louisiana at Lafayette. His current research interests include electronics, VLSI, microelectronics, random number generators, and spiking neurons.

Sonia Akter (Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical and electronics engineering from American International University-Bangladesh in 2013 and 2015, respectively, and the M.Sc. degree in computer engineering from the University of Louisiana at Lafayette, where she is currently pursuing the Ph.D. degree in computer engineering. Her research interests include hardware security, cryptography, VLSI, microelectronics, random number generators, 3D-IC, and the IoT.

Magdy Bayoumi (Life Fellow, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from Cairo University, Giza, Egypt, in 1973 and 1977, respectively, the M.Sc. degree in computer engineering from Washington University in St. Louis, St. Louis, MO, USA, in 1980, and the Ph.D. degree in electrical engineering from the University of Windsor, Windsor, ON, Canada, in 1984.

He is currently the Department Head of the Electrical and Computer Engineering Department, University of Louisiana at Lafayette, Lafayette, LA, USA. His current research interests include very-large-scale integration (VLSI) design and architectures, digital signal processing (DSP), and wireless ad hoc and sensor networks.



Kasem Khalil (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from Assiut University, Asyut, Egypt, in 2009 and 2014, respectively, and the Ph.D. degree in computer engineering from the Center of Advanced Computer Studies (CACS), University of Louisiana at Lafayette, Lafayette, LA, USA, in 2021.

His research interests include electronics, very large-scale integration (VLSI), microelectronics, reconfigurable hardware, self-healing hardware systems, machine learning, hardware accelerators, network-on-chip, artificial intelligence, hardware security, cryptography, intelligent hardware systems, and the Internet of Things. He was a recipient of IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS Prize Paper Award (IEEE Circuits and Systems Society VLSI Paper Award) in 2023. He has also been recognized in the Stanford/Elsevier World's Top 2% Scientists for 2024 and 2025. He has been serving as an Associate Editor for *Microelectronics* journal (Elsevier) since 2022 and as a Guest Editor for the MDPI Sensors and Electronics journals.